

Machine Learning in Business

MIS710 – A2

Technical Report

Data2Intel – Consulting Project



Student Name: Quang Huy Le

Student ID: 224279168

Table of Contents

<i>Excutive Summary</i>	3
<i>I. Introduction</i>	3
<i>II. Approach</i>	3
<i>III. Data Preparation and Exploratory Data Analysis</i>	3
1. Data Understanding and Initial Data Preparation	3
2. EDA	4
<i>Univariate</i>	4
<i>Bivariate</i>	6
<i>Multivariate</i>	9
3. Feature Selection, Data Processing and Data Splitting	9
<i>IV. Model Development and Evaluation</i>	11
1. Supervised Machine Learning	11
Model Selection Rationale:.....	11
2. Unsupervised Machine Learning – Clustering using k-Means.....	13
<i>V. Solution Recommedation</i>	15
1. k-fold Cross Validation and Solution Recommedation	15
2. Future Engagement.....	16
<i>VI. Technical recommendation</i>	16
1. Development and Testing Environment.....	16
2. Machine Process Diagram.....	16
3. Suggestions for Maintenance	17

Executive Summary

This project focused on predicting student performance and identifying those at risk in Year 3 writing assessments using logistic regression, k-nearest neighbors (k-NN), and k-means clustering. After cleaning and processing a dataset of 2,000 students, logistic regression emerged as the best-performing model with 74.75% accuracy and strong precision (65.68%), while k-NN showed higher recall but lower precision. Clustering provided additional insights for targeted interventions that literacy and numeracy skills should be improved to lower the writing risk. The project aimed to help educators identify at-risk students early, allowing for timely support and resource allocation.

Logistic regression is recommended for deployment due to its balance of accuracy and precision. To maintain performance, the model should be retrained regularly with updated data, and future efforts should focus on improving recall. Collaboration with educators is crucial for refining the model's effectiveness, ensuring it adapts to evolving student needs. Continuous monitoring will ensure sustained success in improving literacy outcomes.

I. Introduction

The primary objective of this project is to address the increasing number of students at risk of underperforming in Year 3 writing assessments. By utilizing machine learning models, the project aims to predict which students are most likely to struggle, enabling the school to provide timely and targeted interventions. In the broader business context, improving student performance not only enhances individual academic outcomes but also strengthens the school's overall reputation and success. The value proposition lies in using data-driven insights to improve educational support and ensure that all students receive the necessary resources to succeed.

II. Approach

This project will utilize two key types of machine learning approaches:

- **Supervised Machine Learning:** Logistic regression and k-NN models will be employed. Logistic regression is used to predict the likelihood of students being at risk based on given features, offering interpretable results about risk factors. The k-NN model, a classification algorithm, will identify at-risk students by comparing them to similar students based on proximity in feature space.
- **Unsupervised Machine Learning:** Clustering via k-Means will group students into distinct clusters, helping to segment the student population based on shared characteristics, which provides deeper insights for tailored interventions.

The primary objective is to determine the best predictive model for identifying at-risk students and effectively segment the population into clusters based on performance and related attributes.

III. Data Preparation and Exploratory Data Analysis

1. Data Understanding and Initial Data Preparation

The dataset used in this project is provided by Data2Intel including 2000 unique students, with 34 attributes covering information on students' demographics, school characteristics, family background, and students' performance on literacy and numeracy assessments, focusing on their Year 1 and Year 2. Notably, there is no data in the Year of 2019, which requires further examination.

Data cleansing

The dataset is relatively small with only 2000 observations, containing some anomalies such as:

- Negative values in attributes like Clay-01-SOY, Clay-02-EOY, TextLevel-01-SOY and HRSIW-01-SOY: These negative values are replaced by 0.
- Removed non-disabled student receiving NCCD funding.
- Corrected outlier in NumAbvYear9 with median value.

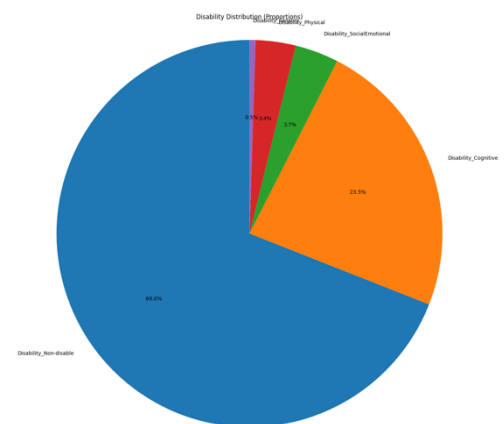
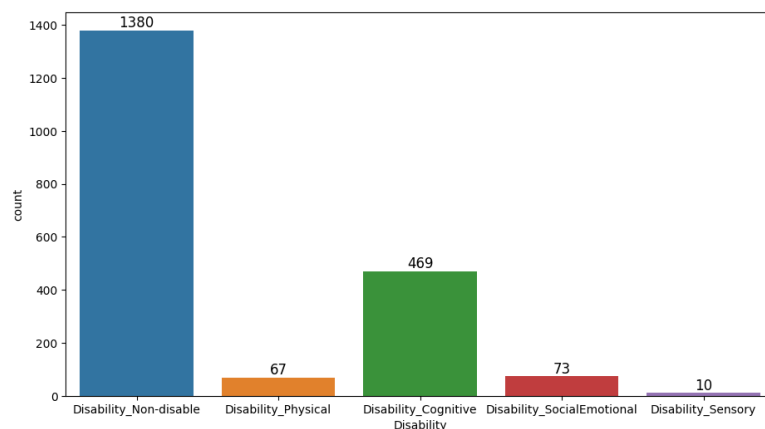
Pre-processing

For further steps in developing predictive model

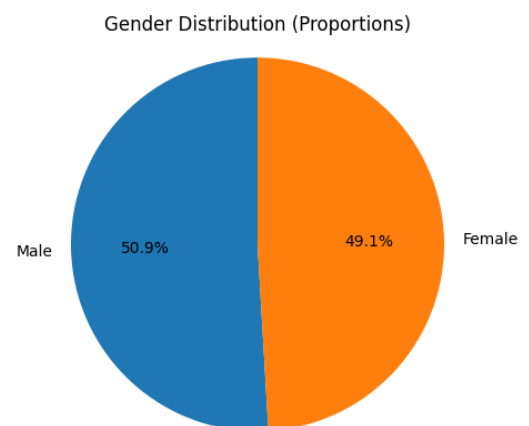
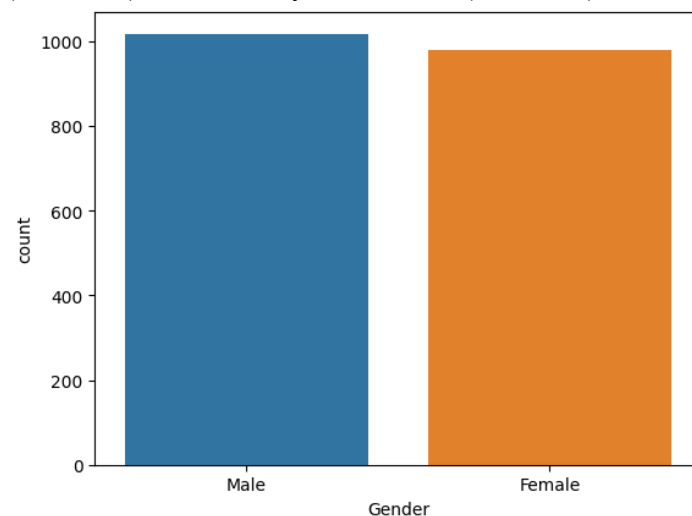
- Convert categorical into numerical using dummy encoding for Gender, Disability, Year3_Writing_At_Risk.
- Removed StudentID column

2. EDA

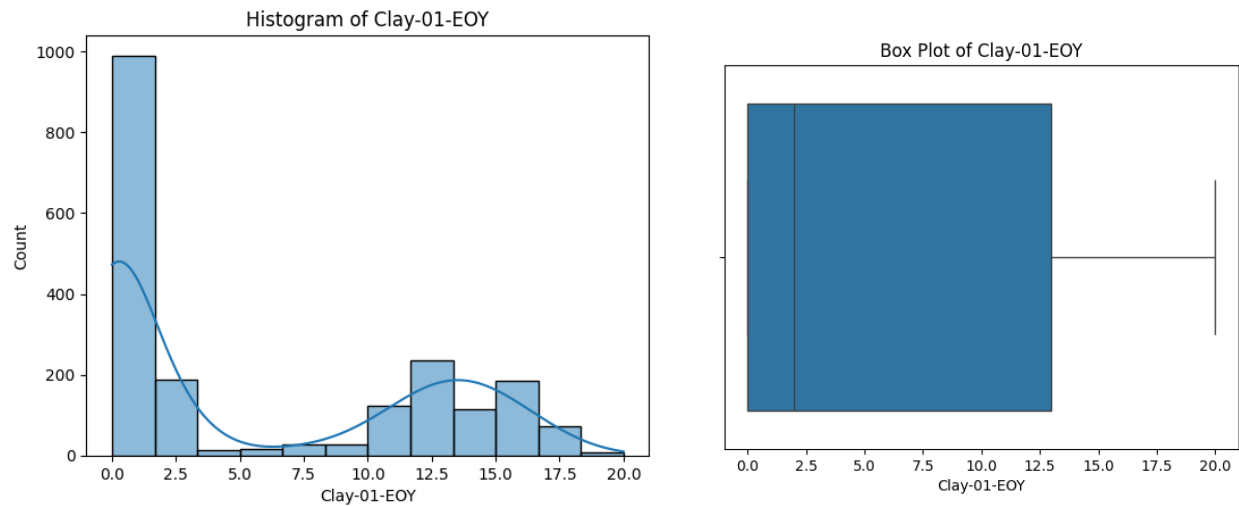
Univariate



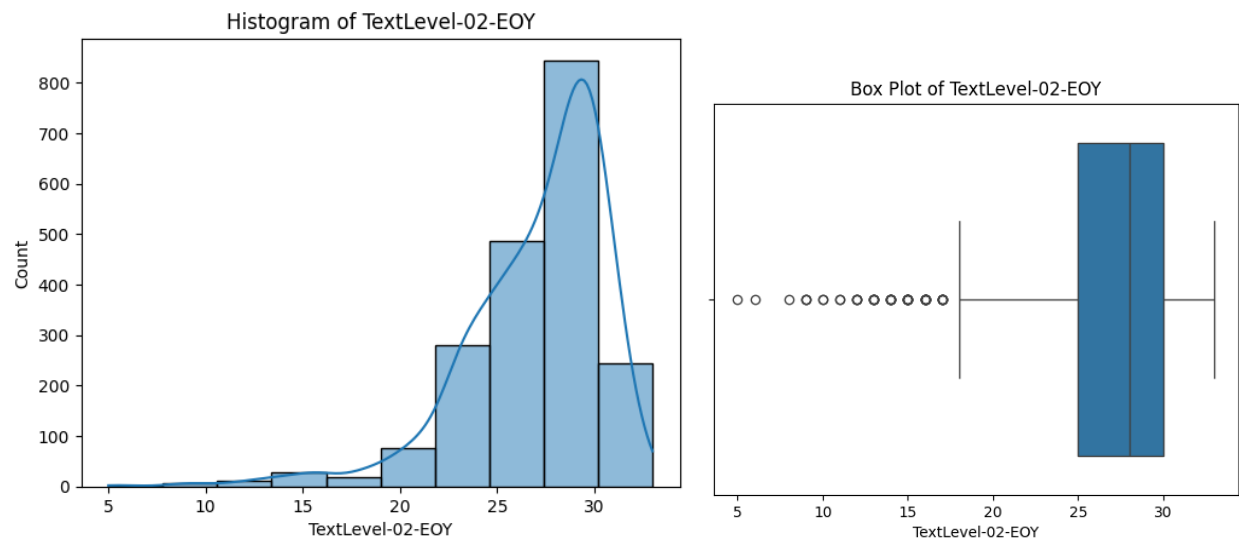
Most students (69%, or 1,380) are non-disabled, while 23.5% (469) have cognitive disabilities. Smaller groups include students with social-emotional disabilities (3.7%, 73), physical disabilities (3.4%, 67), and sensory disabilities (0.5%, 10).



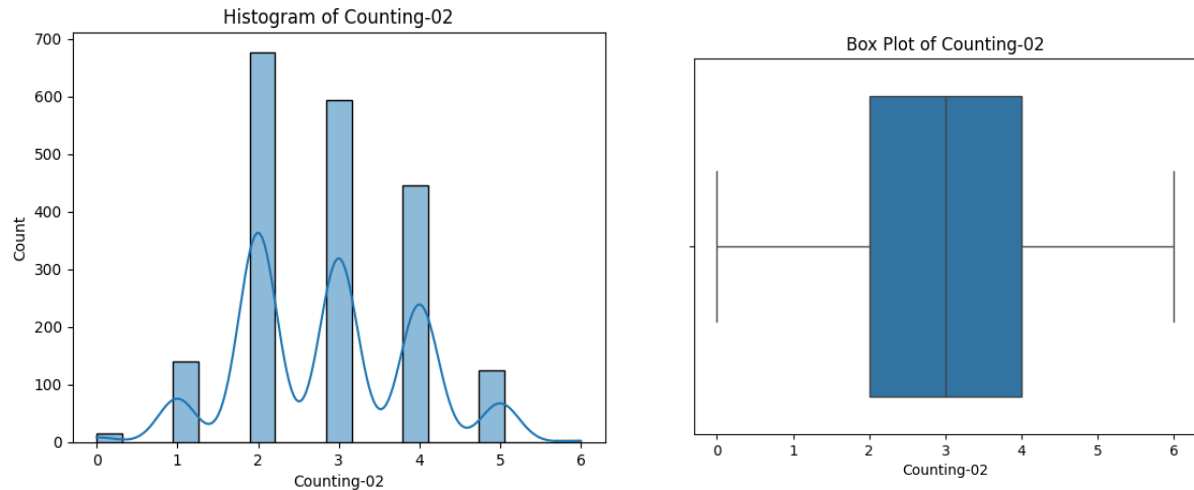
The gender distribution is relatively balanced, with 1,018 males (50.9%) and 981 females (49.1%). This minimal gender disparity help reduce the risk of bias in further analysis and recommendations.



Most students scored between 0 and 12.5. The histogram shows that few students achieved higher scores, as indicated by a right-skewed distribution. The box plot shows a median of around 2.5 and several high score above 12.5, suggesting that while most students scored low, a small group performed exceptionally well.

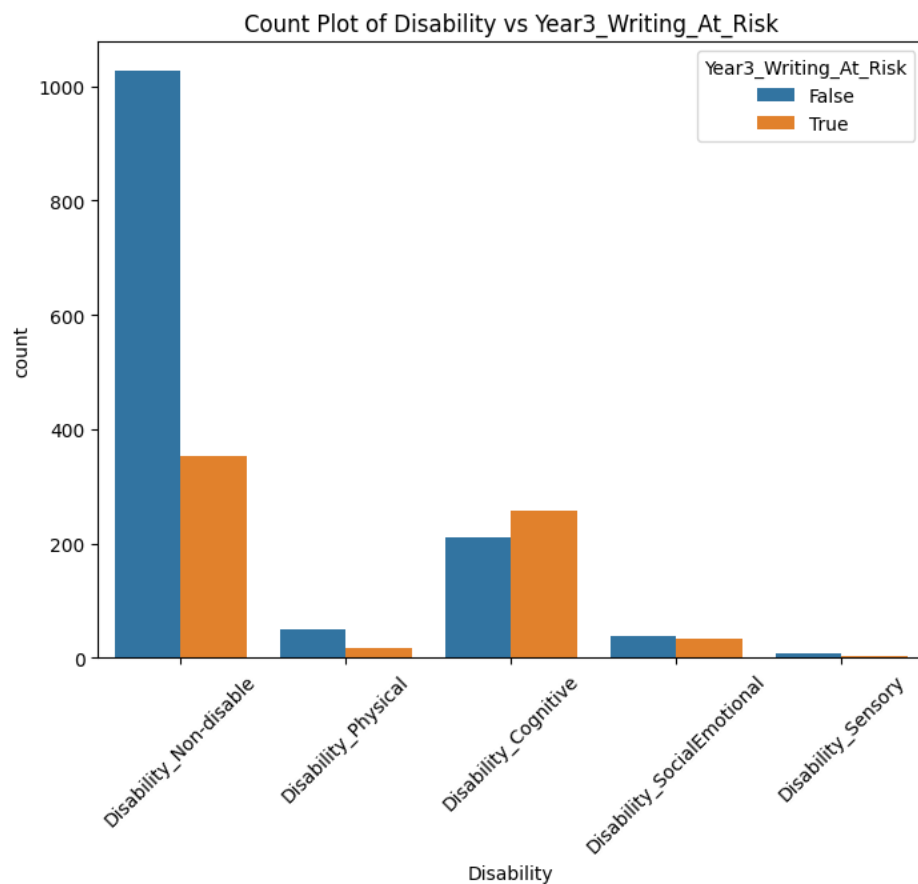


The majority of students scored between 20 and 30, with the peak around 27-30. The distribution is slightly left-skewed, indicating most students performed well in this assessment. However, notable outliers suggest a small group of students significantly underperformed.

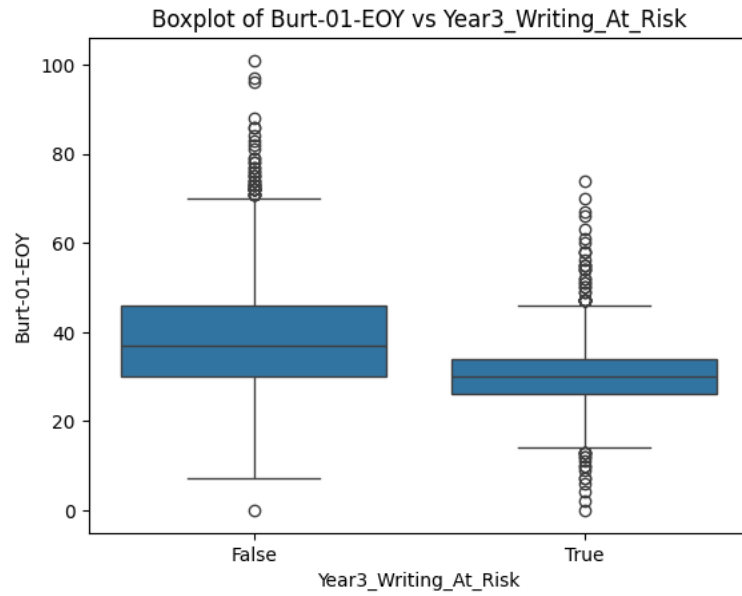


The histogram indicates that most students scored between 2 and 4, with noticeable peaks at 2 and 3. The box plot shows median around 3, with a relatively even distribution of scores across students, indicating overall consistent performance.

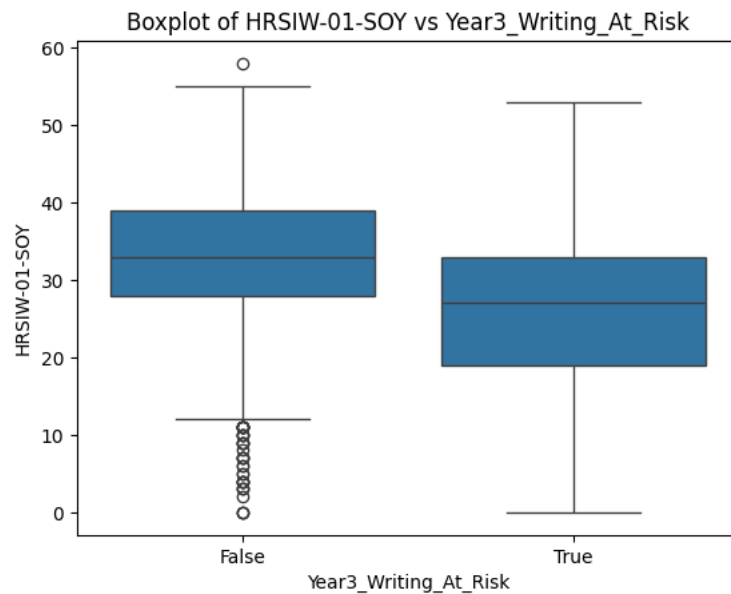
Bivariate



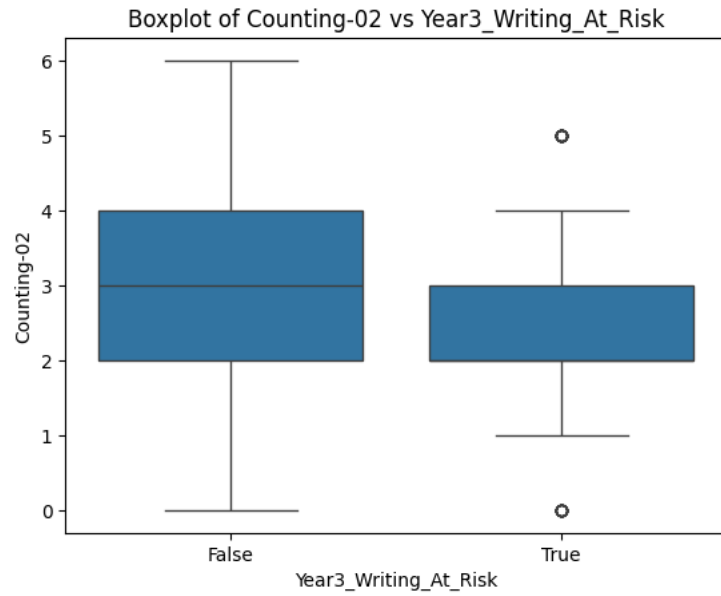
The majority of students without disabilities are not at risk. For students with cognitive disabilities, a higher proportion is at risk, indicating that cognitive disabilities are a significant predictor of writing risk.



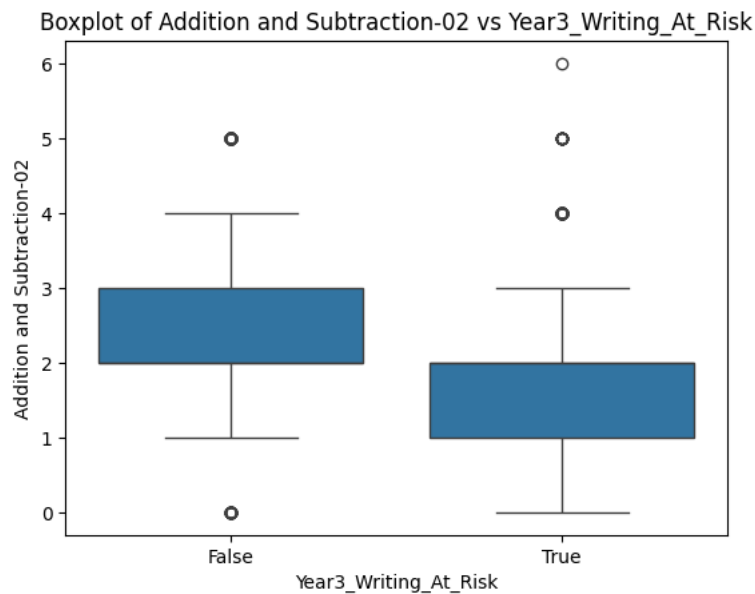
Students at risk have lower Burt-01-EOY scores compared to those not at risk. The median score for at-risk students is significantly lower, around 30 compared to nearly 40 for their counterparts. Burt-01-EOY may correlate with writing performance.



Students at risk tend to have lower HRSIW-01-SOY scores comparing to their counterparts. There's a visible difference in median values, confirming the predictive power of HRSIW-01-SOY in assessing writing risk.

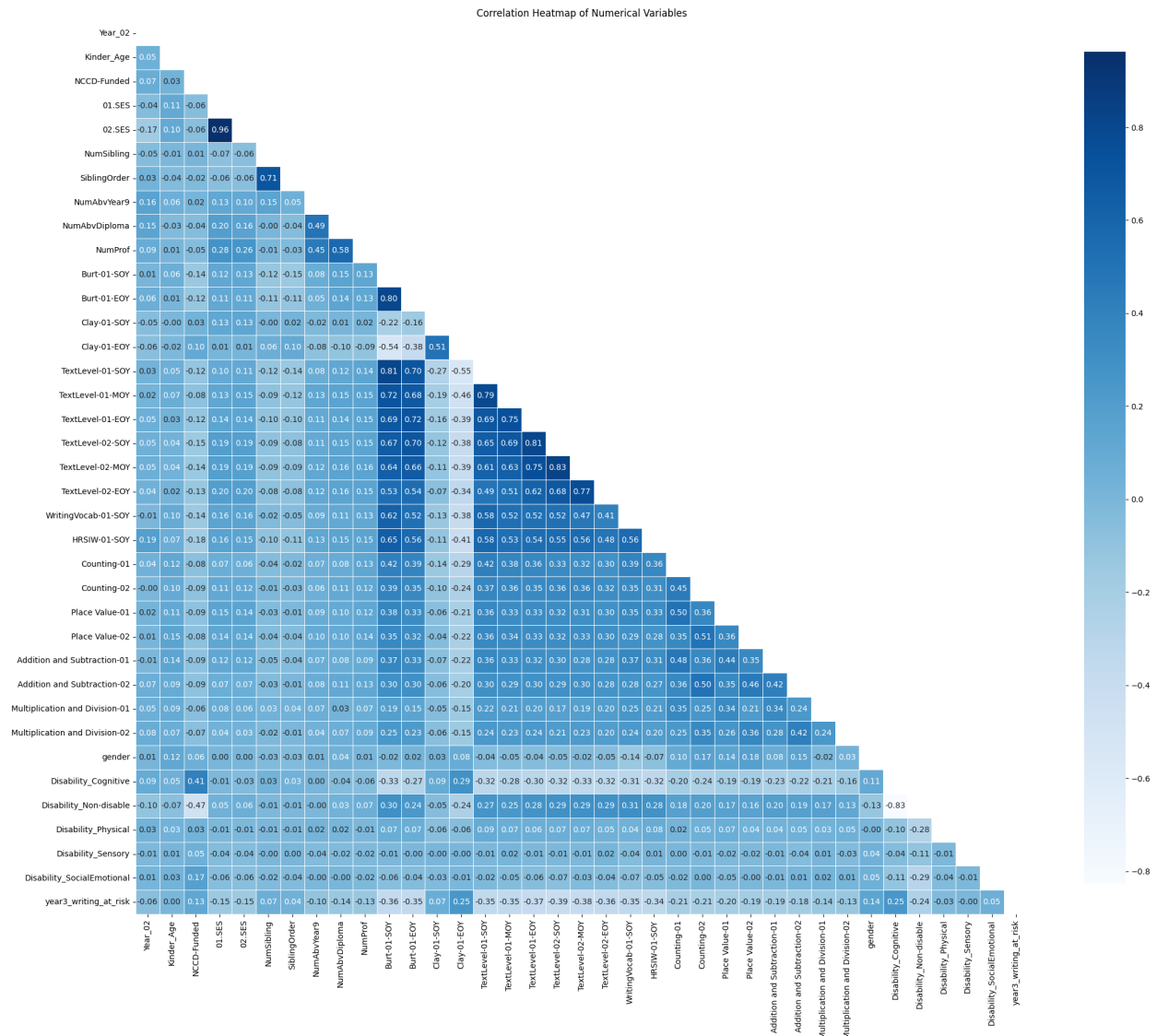


Students not at risk shows a higher median of Counting-02 scores, around 3, with many students scoring between 2 and 4. Students at risk demonstrates a lower median of Counting-02 scores, closer to 2. Some outliers perform notably poorly, around 0.



Students not at risk have a higher median score in Addition and Subtraction-02, around 3, with outlier scores as high as 5. Students at risk show lower median scores, around 2. Notably, there are outliers perform well, at 5 and 6, which requires more examination.

Multivariate



The correlation heatmap show that most of the literacy and numeracy skills demonstrate weak to moderate negative relationship with year3_writing_at_risk, with the highest correlation is TextLevel-02-SOY (-0.39) and the lowest is Clay-01-SOY (0.07).

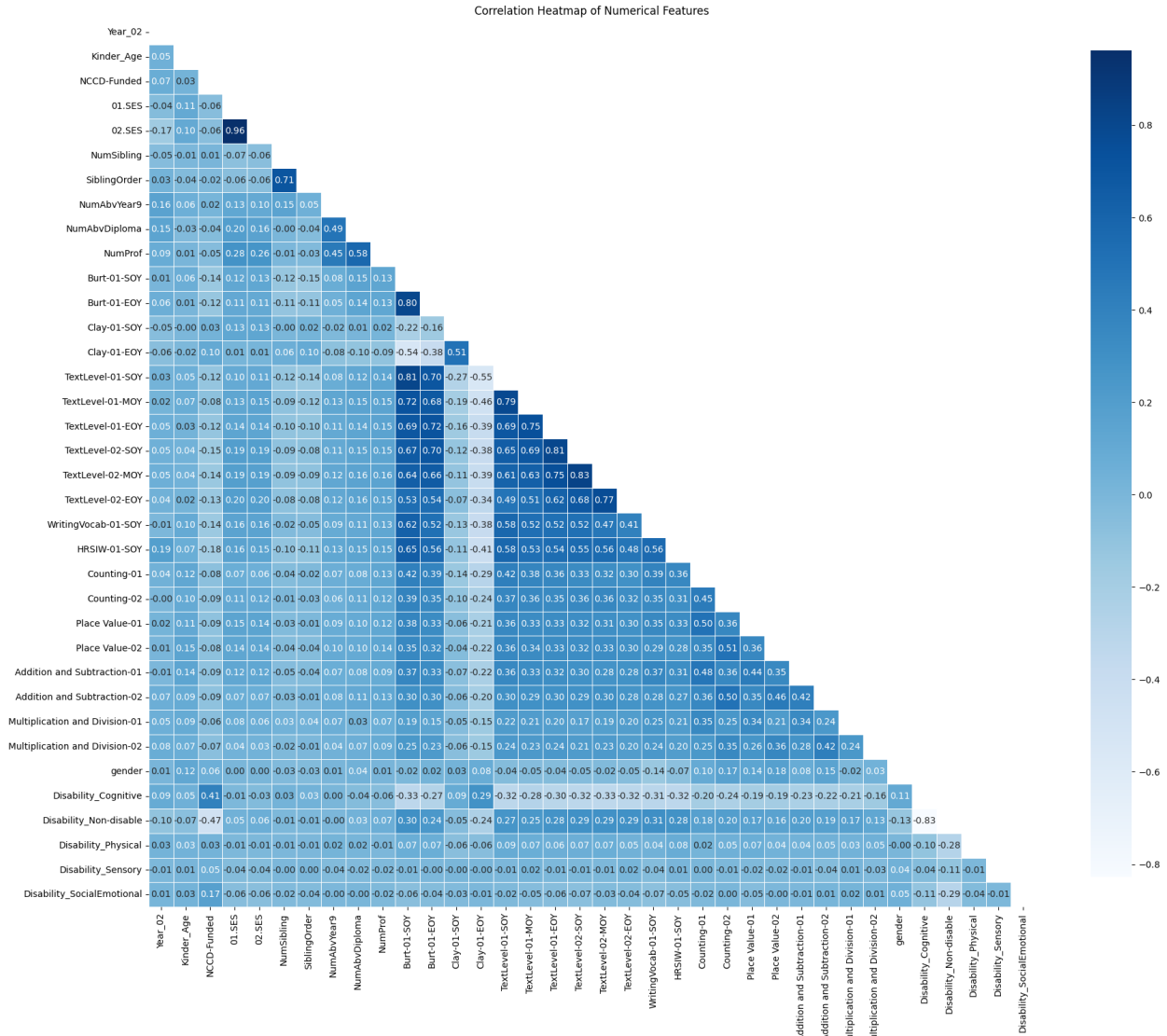
School characteristics and family background show no relationship with target variables. For student demographics, only Disability_Non-disable and Disability_Cognitive show moderate relationship with year3_writing_at_risk.

3. Feature Selection, Data Processing and Data Splitting

Feature Selection for Logistic Regression Model

Based on the correlation heatmap demonstrated above, selected attributes with correlation ≥ 0.2 or ≤ -0.2 with Year3_Writing_At_Risk. Excluded features with high multicollinearity (correlation ≥ 0.8 or ≤ -0.8). Chosen features: Burt-01-SOY, Clay-01-EOY, TextLevel-01-MOY, TextLevel-02-SOY, TextLevel-02-EOY, WritingVocab-01-SOY, HRSIW-01-SOY, Counting-01, Counting-02, Place Value-01, Disability_Cognitive.

Feature Selection for Clustering



Excluded attributes with multicollinearity (correlation ≥ 0.7 or ≤ -0.7). Selected features based on variance > 1 : Year_02, 02.SES, Burt-01-EOY, Clay-01-SOY, Clay-01-EOY, TextLevel-02-EOY, WritingVocab-01-SOY, HRSIW-01-SOY, Counting-02, Addition and Subtraction-02 are chosen.

Data Scaling

For k-NN and Clustering using k-Means, numerical data need to be scaled into the same range as both k-NN and k-means clustering rely on distance measures.

Data Splitting

The dataset was divided into two portions: 80% was allocated for training the model, while the remaining 20% was set aside for testing. This approach ensures the model is trained on a substantial and varied subset of the data, helping it learn a broad spectrum of patterns and relationships. The 20% reserved for testing offers an impartial assessment of the model's performance, giving insight into its ability to generalize and accurately predict ratings on new, unseen data.

IV. Model Development and Evaluation

1. Supervised Machine Learning

Model Selection Rationale:

Logistic Regression:

- Suitable for binary classification ("Year3_Writing_At_Risk")
- Highly interpretable, providing clear feature contribution insights
- Effective with small datasets (2000 observations)

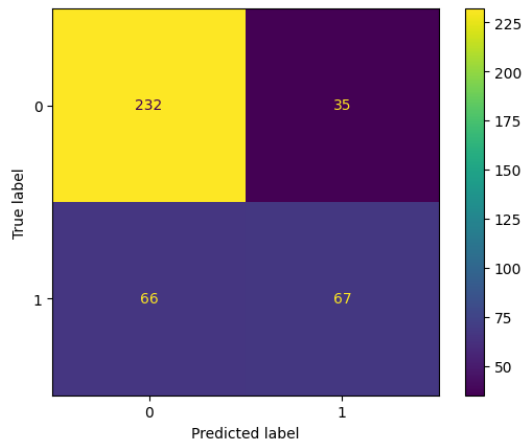
k-NN:

- Non-parametric, instance-based learning
- Handles diverse data distributions without extensive preprocessing
- Predicts based on similar data points, useful for identifying at-risk students

These models were chosen for their suitability to the dataset characteristics and the binary classification problem at hand.

A. Performance Metrics

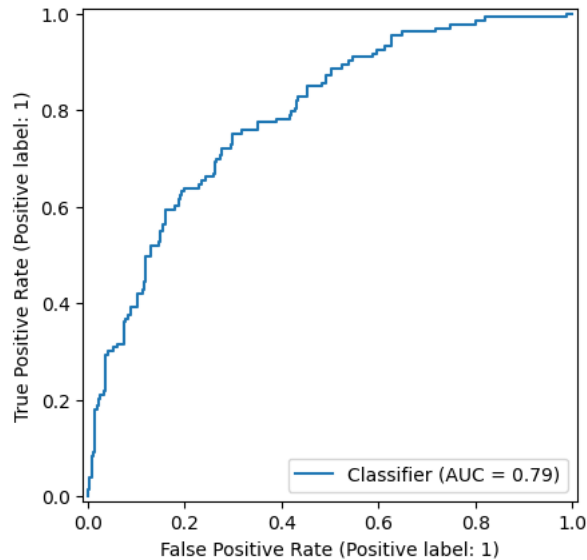
Logistic Regression



The model correctly identify 232 true negatives and 67 true positives. However, it also generated 66 false negatives (students at risk but not flagged) and 35 false positives (incorrectly flagged as at risk).

Accuracy	Precision	Recall	F1
0.7475	0.6568	0.5037	0.5702

- Accuracy (74.75%): The model correctly classifies 74.75% of all cases. At 74.75%, the model's overall classification is reasonably good.
- Precision (65.68%): When the model predicts a student is at risk, it's correct 65.68% of the time. The model is moderately reliable.
- Recall (50.37%): The model identifies 50.37% of the actual at-risk students, showing that the model misses half of the actual at-risk students, which is concerning.
- F1 Score (57.02%): A balance between precision and recall. The moderate F1 suggests that there is room for improving both precision and recall.



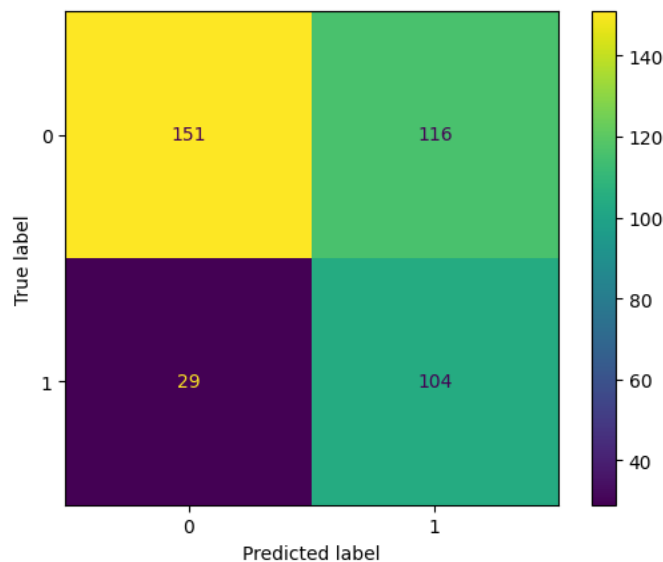
The ROC curve shows a trade-off between sensitivity and false positives, with an AUC of 0.79, suggesting the model has good discrimination, though it can be improved.

Equation:

$$\text{Year3_Writing_At_Risk} = 4.068 - 0.006878 \cdot \text{Burt-01-SOY} + 0.000942 \cdot \text{Clay-01-EOY} - 0.046593 \cdot \text{TextLevel-01-MOY} - 0.076131 \cdot \text{TextLevel-02-SOY} - 0.053802 \cdot \text{TextLevel-02-EOY} - 0.028571 \cdot \text{WritingVocab-01-SOY} - 0.009934 \cdot \text{HRSIW-01-SOY} - 0.007694 \cdot \text{Counting-01} - 0.030744 \cdot \text{Counting-02} - 0.048489 \cdot \text{Place Value-01} + 0.379213 \cdot \text{Disability_Cognitive}$$

- Positive coefficients (Disability_Cognitive has highest) suggest that higher values of these predictors increase the likelihood of a student being at risk in Year 3 writing.
- Negative coefficients (TextLevel-02-SOY has the highest) indicate that higher values of these predictors reduce the likelihood of being at risk.

Optimised k -NN Model with $k = 13$

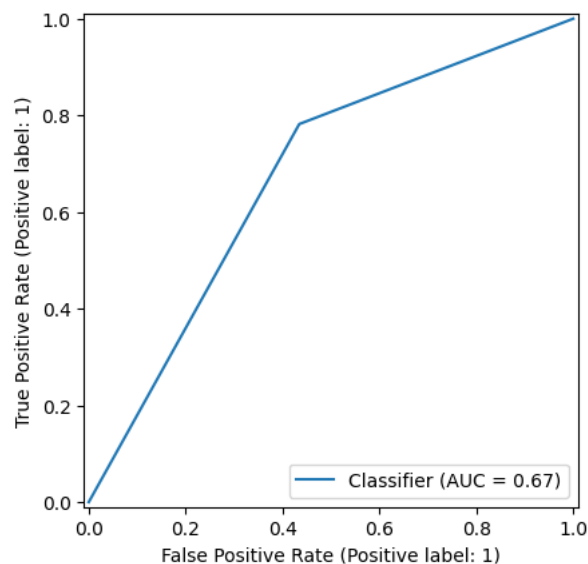


The model correctly identified 151 true negatives and 104 true positives but also had 116 false positives and 29 false negatives, indicating difficulties in correctly identifying at-risk students.

Accuracy	Precision	Recall	F1
----------	-----------	--------	----

0.6375	0.4727	0.7819	0.5892
--------	--------	--------	--------

- Accuracy of 63.75% suggests that a moderate accuracy rate, indicating the model correctly classifies 63.75% of the samples.
- Precision of 47.27%, meaning less than half of the at-risk predictions were accurate.
- Recall of 78.19%, showing that the model is relatively good at identifying most of the actual at-risk students.
- F1 Score of 58.92%, which balances precision and recall, is moderate, indicating the model has an overall fair performance.



The AUC of 0.67 suggests the model is only slightly better than random guessing and has room for improvement in distinguishing between at-risk and not-at-risk students.

B. Model Comparison

	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.7475	0.6568	0.5037	0.5702	0.79
k-NN	0.6375	0.4727	0.7819	0.5892	0.67

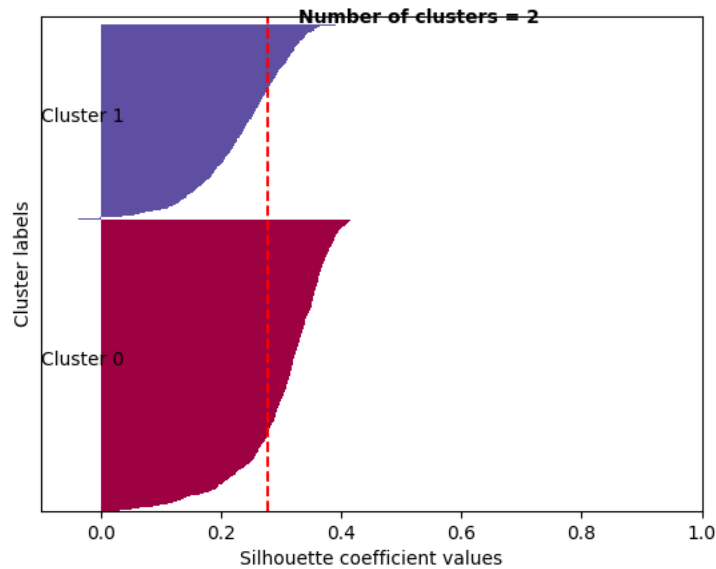
- Accuracy: Logistic regression is higher at 0.7475, compared to k-NN's 0.6375.
- Precision: Logistic regression performs better with 0.6568, while k-NN achieves 0.4727.
- Recall: k-NN is higher with 0.7819, compared to logistic regression's 0.5037.
- F1 Score: k-NN has a slight advantage at 0.5892, while logistic regression scores 0.5702.
- AUC: Logistic regression leads with 0.79, surpassing k-NN's 0.67.

Logistic Regression has higher accuracy, precision, and AUC, making it a better choice for predicting at-risk students with fewer false positives. k-NN, while better in terms of recall and F1 score, suffers from lower precision and accuracy, making it more likely to classify students at-risk incorrectly.

2. Unsupervised Machine Learning – Clustering using k-Means

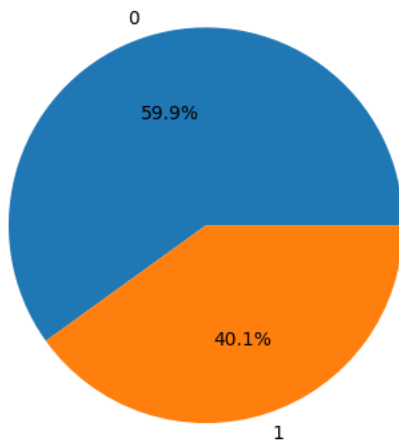
K-Means clustering was selected because it can reveal patterns in student performance without the need for predefined categories. It is particularly useful for grouping students based on similarities in their performance metrics.

With the optimised clustering model with $k = 2$, the model performance output as follows:

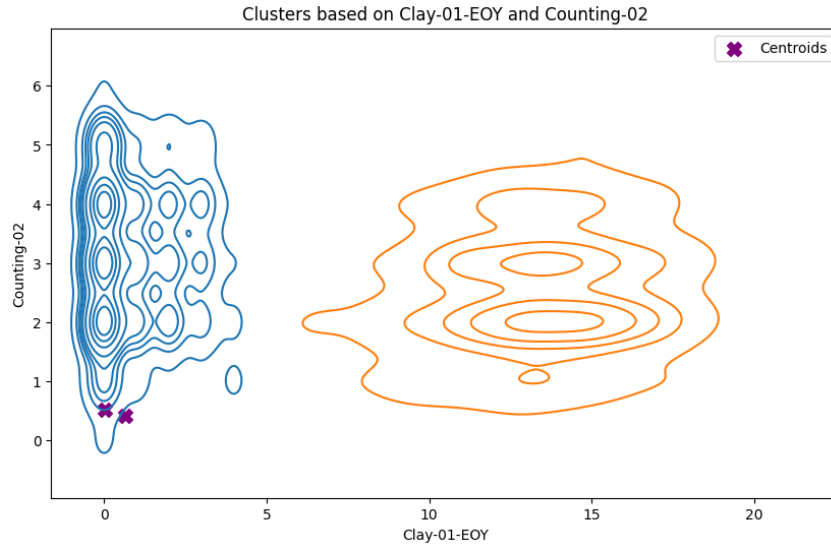


- WCSS (661.991): This indicates moderate cohesion within clusters, showing the total squared distance between each data point and its cluster centroid.
- Davies-Bouldin Index (1.532): The score reflects overlapping clusters, suggesting that they are not perfectly separated and could be improved.
- Silhouette Score (0.278): This low score suggests weak cluster definition with moderate overlap, indicating room for better separation of clusters.

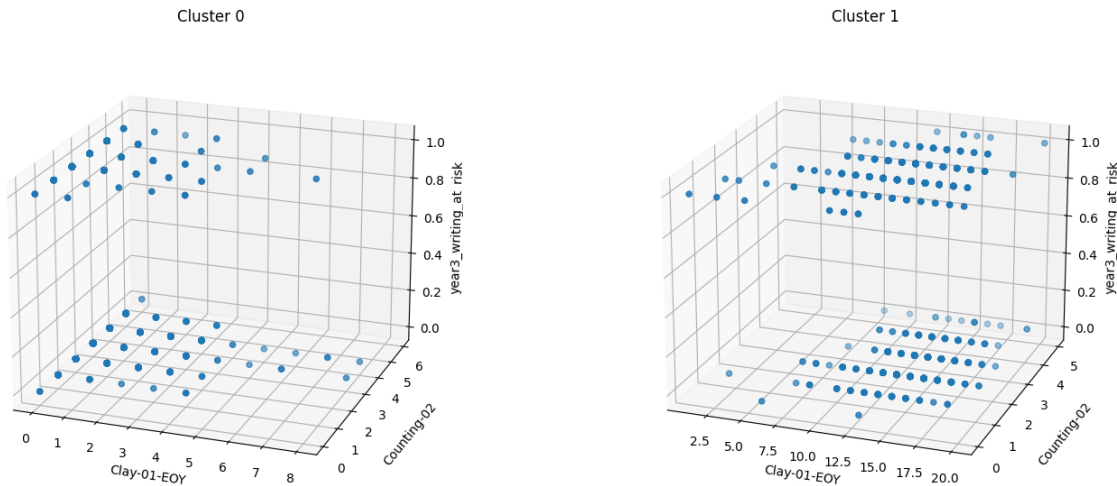
Distribution of Clusters



The K-means clustering results divide the students into two main clusters based on their Clay-01-EOY and Counting-02 scores. Cluster 0 (59.9%) represents students with lower scores in both features, while Cluster 1 (40.1%) groups students with higher scores.



Cluster 0 groups students with lower Clay-01-EOY and Counting-02 scores, indicating they may need additional support in numeracy and literacy. Cluster 1 includes students with moderate to high scores in both areas, suggesting better academic performance.



Cluster 0, as shown in the 3D scatter plot, shows a higher concentration of students at risk in Year 3 writing, while Cluster 1 has fewer students at risk, given their stronger performance in numeracy and literacy metrics.

V. Solution Recommendation

1. *k*-fold Cross Validation and Solution Recommendation

A 10-fold cross-validation was performed for both logistic regression and k-NN ($k = 13$). The performance metrics are shown below:

	Accuracy	Precision	Recall	F1
Logistic Regression	0.733 (+/- 0.011)	0.521 (+/- 0.029)	0.648 (+/- 0.023)	0.437 (+/- 0.035)
k-NN	0.715 (+/- 0.020)	0.606 (+/- 0.044)	0.426 (+/- 0.033)	0.499 (+/- 0.033)

Both model show slight variability, therefore, they are both reliable. However, as the purpose of this project is to indentify at-risk student, with higher accuracy and recall, logistic regression is chosen.

2. Future Engagement

To maintain and enhance the results, future efforts should focus on continuous collaboration with the school and students to obtain updated and more diverse data. This will help improve model accuracy and ensure the dataset reflects changes in student performance and allow for model retraining and fine-tuning, ensuring its long-term reliability.

Additionally, detailed deployment plans and integration into the school's system should be made, with ongoing monitoring and updates based on new data to maintain effectiveness and alignment with educational needs.

VI. Technical recommendation

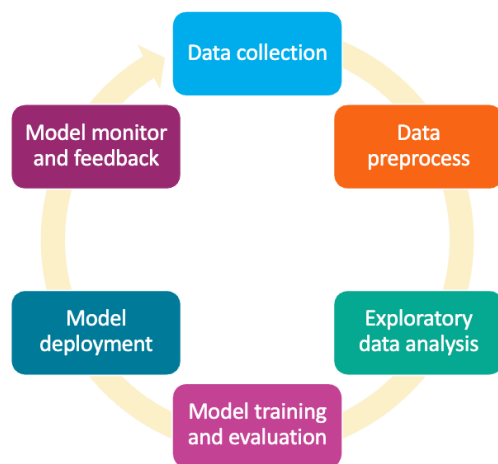
1. Development and Testing Environment

The model development and testing were carried out using Python, a popular language for machine learning and data science. The following software libraries were used:

- pandas and numpy for data manipulation and numerical analysis.
- seaborn and matplotlib.pyplot for data visualization.
- itertools for efficient looping and handling of iterators.
- sklearn for implementing machine learning algorithms and evaluation metrics.

The computing environment was Google Colab, which provided a cloud-based, collaborative platform with access to GPU support, ensuring smooth execution and scalability for future updates and model deployment.

2. Machine Process Diagram



The diagram show the cycle of a machine learning:

- Data Collection: Gathering student data to feed into the machine learning model.
- Data Preprocessing: Cleaning, replacing negative value with 0, removing inconsistencies of disability and fund, replacing anomalies of NumAbvYear9 with median value and scaling numerical data to ensure it's ready for analysis.

- Exploratory Data Analysis (EDA): Analyzing the data through univariate, bivariate and multivariate to understand patterns, relationships, and insights that inform model development.
- Model Training and Evaluation: Building logistic regression, k-NN and clustering using k-Means model and fine-tuning the machine learning model, followed by cross-validation and evaluation using metrics like accuracy, precision, recall and f1.
- Model Deployment: Implementing the trained model in a real-world environment where it can make predictions.
- Model Monitoring and Feedback: Continuously tracking the model's performance in production and refining it based on feedback to maintain its effectiveness.

This cycle is iterative and helps in continuous improvement of machine learning solutions.

3. Suggestions for Maintenance

To maintain accuracy and relevance over time:

- Update Data: Continuously working with school and gather new data to reflect current trends of student performance.
- Retrain Model: Regularly retrain the model to adapt to changes in student performance.
- Monitor Performance: Track model metrics (accuracy, precision, etc.) to detect performance drops.
- Integrate Feedback: Incorporate feedback from educators to ensure the model remains practical.
- Reassess Features: Regularly evaluate and update input features based on evolving educational factors.